# Supplementary Material

Overlapping construction plus a position-blind split:
a curation defect in genomic sequence benchmarks and the model rankings it inverts

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Section numbers of the form §3.6 refer to the main text; §S1, §S2, ... refer to this document. Every number here is re-derived from the CSV files released with the code, and the gate script `audit/tools/check_numbers.py` checks both documents against those files.

## Contents

# S1 Extended methods

## S1.1 Datasets

Table S1: Datasets in the Genomic Benchmarks binary suite, and the optional three-class set. Lengths are the median with min–max range where variable; class balance is the positive-class fraction. All balanced datasets are curated to 0.500 in both train and test, so balance is preprocessed rather than natural.

| Dataset | $n$ (full) | Length (bp) | Class balance | Test fraction |
|---|---|---|---|---|
| human_nontata_promoters | 36,131 | 251 | 0.544 | 0.25 |
| human_enhancers_ensembl | 154,842 | 269 (2–573) | 0.500 | 0.20 |
| human_ocr_ensembl | 174,756 | 315 (71–593) | 0.500 | 0.20 |
| human_enhancers_cohn | 27,791 | 500 | 0.500 | 0.25 |
| demo_coding_vs_intergenomic_seqs | 100,000 | 200 | 0.500 | 0.25 |
| demo_human_or_worm | 100,000 | 200 | 0.500 | 0.25 |
| drosophila_enhancers_stark | 6,914 | 2142 (236–3237) | 0.500 | 0.25 |
| human_ensembl_regulatory (3-class) | 289,061 | 401 (89–802) | 3 classes | 0.20 |

## S1.2 Model roster: full specification

The four-model comparison used for the report card is: logistic regression (`C=1.0`, `max_iter=5000`); a linear support-vector machine (`LinearSVC`, `C=1.0`, `dual=False`, `max_iter=5000`); a random forest (`n_estimators=150`, scikit-learn defaults `min_samples_leaf=1`, `max_depth=None`); and gradient-boosted trees (`HistGradientBoosting`, library defaults). The nine-model roster reuses those four *verbatim*—same objects, same normalization, same seeds—and adds: 1- and 15-nearest neighbours; extremely randomized trees (`n_estimators=150`, defaults); a multilayer perceptron (one hidden layer of 128 units, `early_stopping=True`, `max_iter=60`); and Gaussian naive Bayes. The model seed is 0 throughout.

The roster spans memorization propensity end to end rather than accuracy. 1-nearest-neighbour is the definitional memorizer—on a near-duplicate its prediction simply *is* the training label—so it bounds the axis from above; Gaussian naive Bayes bounds it from below. 15-nearest-neighbour was designated memorization-prone in advance and turns out to sit at the bottom of the corrected-gap ordering, because averaging over fifteen neighbours is what stops a nearest-neighbour rule from memorizing; we keep it in the memorization-prone set as registered rather than reclassifying it after the fact.

**Normalization policy.** L2 normalization is applied via a `Normalizer` pipeline step to the scale-sensitive learners only: both linear models, both nearest-neighbour rules and the MLP. The tree ensembles consume raw $k$-mer counts, for which monotone feature scaling is irrelevant, and Gaussian naive Bayes standardizes internally.

**Feature settings.** Overlapping $k$-mer counts at $k \in \{4, 6\}$; main-text results use $k = 6$ unless stated. At $k = 4$ the demotion reproduces, with full-scale random-forest drops of 0.153 and 0.084 on the two leaky datasets and the demotion holding under accuracy and AUROC alike. The three-class row of the report card is a $k = 4$ figure.

The nine-model roster itself is Table 3 of the main text, which carries, per model, both forms of the graded memorization gap, the as-shipped and corrected accuracies, the drop with its two-arm cluster-bootstrap interval, and the rank on each split.

**Reading conventions for the roster.** An inversion criterion designed for four models is too permissive for nine, because with a wider roster the top two are often within noise and a "top model changed by more than a point" rule fires on a *clean* control from repartitioning alone; hence the two-arm paired requirement stated in the main text. And the margin between the two models that *swap* is a different quantity from the margin between as-shipped first and second place: the latter can be a statistical tie while the former is large, which is the situation on *human_enhancers_ensembl*.

### S1.3 The prevalence-corrected graded memorization gap

The split-internal graded memorization gap is

$$g_M \;=\; \mathrm{acc}_M[\,s \geq 0.9\,] \;-\; \mathrm{acc}_M[\,s < 0.5\,], \tag{S1}$$

for $s$ the maximum 8-mer Jaccard of a test sequence to any training sequence. Raw, the statistic is confounded by stratum class composition. The $\geq 0.9$ stratum is 99.97% positive on *human_enhancers_ensembl* against 19.68% in the novel stratum, and 0.15% against 98.17% on *human_nontata_promoters*—opposite directions on the two datasets. A constant classifier that always predicts the positive class, and therefore memorizes nothing, scores a raw gap of +0.803 on the first and −0.980 on the second, so any model with a class-prediction bias inherits a large gap of the corresponding sign for free.

We therefore compute the gap from *balanced* accuracy—the unweighted mean of per-class recall—within each stratum. A constant classifier scores 0.5 in every stratum under that definition, hence a corrected gap of exactly zero. Both forms appear side by side in Table S2 so the size of the confound is visible. The correction disposes of two raw values never interpretable as memorization: 15-nearest-neighbour's raw −0.684 and naive Bayes's −0.234 become +0.016 and +0.129, i.e. artifacts of predicting the minority class in a stratum that is 99.97% positive.

Table S2: Nine-model roster on *human_enhancers_ensembl*, full scale, as-shipped versus near-duplicate-aware split. $\mathrm{Gap}_{\mathrm{bal}}$ is the graded memorization gap computed from *balanced* accuracy within each stratum and is the statistic relied on; $\mathrm{Gap}_{\mathrm{raw}}$ is the uncorrected accuracy difference, shown only for comparison because it is confounded by stratum class composition—a constant classifier scores +0.803 on this dataset, which is why the raw column carries two large negative values that do not indicate "negative memorization". The corrected column orders the models as the memorization account predicts and the raw column does not. "Drop" is the accuracy lost under correction, with a two-arm cluster-bootstrap interval; * marks intervals excluding zero. Models are ordered by as-shipped rank. The top three as-shipped finish within 0.0015 of one another—a tie—yet span 23 points on the corrected split, and the corrected winner is the linear SVM. Generated from `results/roster_rankings.csv`; the prevalence-corrected gaps are exploratory (main text §3.7).

| Model | $\mathrm{Gap}_{\mathrm{bal}}$ | $\mathrm{Gap}_{\mathrm{raw}}$ | As-shipped | Corrected | Drop | Rank |
|---|---|---|---|---|---|---|
| MLP | **0.234** | 0.159 | **0.8736** | 0.7684 | 0.105* | 1→3 |
| kNN-1 | **0.461** | 0.208 | 0.8729 | 0.5370 | 0.336* | 2→9 |
| ExtraTrees | **0.370** | 0.210 | 0.8721 | 0.6278 | 0.244* | 3→7 |
| RandomForest | **0.319** | 0.230 | 0.8596 | 0.6957 | 0.164* | 4→5 |
| LinearSVC | **0.144** | 0.037 | 0.7980 | **0.7975** | 0.001 | 5→1 |
| LogisticReg. | **0.154** | 0.037 | 0.7809 | 0.7806 | 0.000 | 6→2 |
| HistGB | **0.162** | 0.034 | 0.7636 | 0.7608 | 0.003 | 7→4 |
| GaussianNB | **0.129** | −0.234 | 0.6482 | 0.6492 | −0.001 | 8→6 |
| kNN-15 | **0.016** | −0.684 | 0.5357 | 0.5371 | −0.001 | 9→8 |

### S1.4 From-scratch 1D CNN: architecture and grid

The probe is a from-scratch 1D residual convolutional network in the Basset (Kelley et al., 2016) / DeepSEA (Zhou and Troyanskaya, 2015) lineage: one-hot nucleotide input (no $k$-mer features), an initial wide convolution, residual blocks with batch normalization and ReLU, global max pooling and a linear head. It trains on each dataset's training split only, so there is no pretraining confound, with early stopping on a held-out slice of that split. The pre-specified grid is dropout $\in \{0.0, 0.3, 0.6\}$ $\times$ weight decay $\in \{0, 10^{-4}, 10^{-3}\}$, nine cells per dataset, plus one unregularized-to-convergence manipulation-check cell that differs from its matched grid cell only in epoch budget.

The pre-registration document `results/deep_preregistration.md` was committed alongside the results it predicts, in the same commit. It is therefore a binding *stated* condition, not an externally timestamped registration, and we grade it accordingly: our own `results/tier1_preregistration.md` sets the standard that an uncommitted file is not a registration and that same-commit artifacts are weak evidence of priority.

### S1.5 Alignment-based validation

To test whether the effect depends on the $k$-mer edge metric we re-cluster the two leaky datasets with MMseqs2 (Steinegger and Söding, 2017) `easy-cluster` (`--min-seq-id 0.7`, `-c 0.8`), feed the external cluster vector into the *same* whole-cluster assignment path, and refit all four models. Two properties must be stated rather than glossed. First, `mmseqs cluster` defaults to `--cluster-mode 0`, a cascaded greedy set-cover, whereas the main text's splitter uses connected components, so the arm varies the linkage rule *alongside* the metric and does not isolate the metric; on *human_nontata_promoters* MMseqs2 returned a *finer* partition (23,758 clusters against 23,312) despite the looser identity criterion, which is a linkage signature rather than a metric one. Second, `--min-seq-id 0.7` (≈70% identity) and an 8-mer Jaccard of 0.7 (≈97–98% identity) are *not* calibrated to one another despite the shared numeral. Because MMseqs2 is itself $k$-mer-seeded we describe the arm as *alignment-scored*, not $k$-mer-independent; it covers the two leaky datasets only, and clean verdicts are corroborated by coordinate provenance instead.

### S1.6 Imbalanced evaluation, tuning selection, and robustness definitions

**Imbalanced evaluation.** Because the balanced datasets are curated, we add a prevalence stress test (random forest, target positive fraction $\pi \in \{0.5, 0.2, 0.1\}$) scored with prevalence-aware metrics: AUPRC against the prevalence baseline, MCC, balanced accuracy, AUROC, and minority-class precision, recall and $F_1$. Downsampling is applied to both train and test. The near-duplicate-aware splitter is extended with a per-class realized-fraction check and preserves the target prevalence (realized positive fractions 0.5001/0.2007/0.1001 on *human_enhancers_ensembl*). The panel is random-forest-only and the prevalence is induced rather than native, so it does not test the reordering claim under imbalance.

**Tuning selection.** We cross-validate `min_samples_leaf`$\in \{1, 5, 20, 50, 100, 200, 500\}$ for the random forest on the as-shipped *training* set of both leaky datasets, under ordinary stratified 5-fold and under 5-fold in which whole near-duplicate clusters—the same connected components used to build the corrected split—are held out together. The first is what a practitioner would actually run; the second is its leakage-aware counterpart. Each selected model is then scored on both splits. The sweep is random-forest-only.

**Cluster cohesion.** The edge density of a single-linkage near-duplicate component: the fraction of its $\binom{m}{2}$ member pairs that actually exceed the similarity threshold. A component built from genuine mutual near-duplicates has cohesion near 1; one whose members are linked only through intermediates has cohesion near 0, and whole-cluster assignment is then a coarse instrument, because it moves sequences that are not pairwise near-duplicates of one another. Low cohesion is not by itself evidence of chaining across distinct loci: on *human_nontata_promoters* the 420-member largest component is a single 1,421 bp locus tiled by 251 bp windows at a median 2 bp step, and 62% of its member pairs have start offsets of $\geq$ 251 bp and therefore share no sequence at all. That is window geometry. The certification tool returns *leaky-but-correction-unsafe* rather than *leaky* below cohesion 0.5; the cut is calibrated on the two leaky datasets studied here, which fall an order of magnitude either side of it (0.96 and 0.083).

**Design effect.** $1 + (m^* - 1)\rho$ for intraclass correlation $\rho$ and Kish's effective block size $m^* = \sum_i n_i^2/N$ over blocks of sizes $n_i$ summing to $N$. We use $m^*$ rather than the mean block size $\bar{m}$ because these block-size distributions are heavy-tailed—$m^*$ is 4.45 against $\bar{m} = 1.55$ on the corrected *human_nontata_promoters* test set—so the mean-size convention understates the factor more than twofold. Because the design effect is an approximation to a quantity the block bootstrap measures outright, we report it beside that measurement rather than in place of it.

**Novel-only evaluation and the GC control.** A *novel-only* evaluation scores the model trained on the as-shipped split using only test sequences below 0.5 similarity to training, isolating generalization without retraining or re-splitting. The *GC-in-distribution* restriction scores the corrected test set restricted to sequences whose GC content lies inside the interquartile range of the training set's; it discards roughly half of each corrected test set (4,523 of 9,041 and 15,299 of 30,971). The low-complexity check uses `dustmasker` (Morgulis et al., 2006) at defaults to report each sequence's masked fraction.

### S1.7 Pre-registered predictions and interval definitions

Six predictions were committed in code, inside the module that would test each, before that module was run, and all are scored in the Results whether or not they held: P1 1-nearest-neighbour shows the largest graded memorization gap, registered against the *raw* form of that statistic—the prevalence-corrected form is exploratory here and is registered forward for the next dataset, not counted backward (§3.7); P2 it shows the largest split-induced drop; P3 the four memorization-prone learners (1- and 15-nearest neighbours, the forest, extremely randomized trees) outrank the two linear models as shipped and fall below them after correction; P4 on a clean control no model's drop interval excludes zero; P5 cluster-grouped cross-validation selects a more regularized forest than naive cross-validation; P6 a ranking inversion occurs at every imposed leak fraction above the diagnostic threshold of Supplementary §S2 and at none below. Separately, before that suite was examined, we registered for GUE (Zhou et al., 2023) that its short fixed-length human regulatory tasks would be leaky and its multi-species tasks clean (§3.12).

**Intervals and reproducibility.** Because near-duplicates violate the independence a sample-wise bootstrap assumes, we resample whole within-test near-duplicate components as blocks on *both* arms, quantify the design effect using Kish's effective block size rather than the mean (these distributions are heavy-tailed and the mean-size convention understates the factor roughly twofold), report it beside the cluster-versus-sample variance ratio the bootstrap measures directly, cross-check against a Liang–Zeger cluster-robust standard error with the CR2 small-sample correction and a $t$ reference on $G-1$ degrees of freedom (which matters because cluster counts on the corrected arm are modest), and fold the five re-split-seed variance into a combined-source interval; Benjamini–Hochberg at $q = 0.05$ is applied over a declared family of 15 deltas (Benjamini and Hochberg, 1995), whose membership rule is stated rather than left to inference: every per-dataset corrected-split accuracy delta in Table 1 and §3.5, for the two pre-registered model families, on all eight datasets. The cross-suite ranking grid is corrected as its own family of 16, and because two families corrected separately is the more permissive choice we report the pooled sensitivity analysis in §3.13. Two properties of these $p$-values are worth stating: they are recovered from percentile intervals under a normal reference, SE $\approx$ (hi $-$ lo)$/2z$, because the bootstrap draws were not retained, and they are therefore adequate for a verdict question only where deltas sit far from the cut—which is why any delta landing within a factor of two of $q$ is flagged as unresolvable rather than called. Reported 95% CIs are percentile bootstrap intervals over 10,000 draws with the resampling unit stated per number; the three combined-source intervals are instead Wald, $\delta \pm t\sqrt{s^2_{\text{boot}} + s^2_{\text{seed}}}$, with the critical value from a Welch–Satterthwaite effective degrees of freedom rather than the normal quantile, since the seed component carries only four degrees of freedom while the bootstrap component carries hundreds, and labelled where they appear. Two robustness checks on the interval method leave every conclusion unchanged, so we state their agreement rather than adopt them as corrections. A bias-corrected and accelerated (BCa) interval, with the acceleration jackknifed over whole clusters, differs from the percentile interval by at most 0.0006 on every delta. And the Welch correction moves the combined-source intervals negligibly—the effective degrees of freedom are 13 to 114, far from the four a flat $t_4$ would assume—so *human_enhancers_ensembl*'s forest interval is unchanged at $[0.154, 0.174]$ and *human_nontata_promoters*'s shifts only to $[0.087, 0.130]$ (`results/cluster_bootstrap_full.csv`). Models are not refit while the test set is resampled, so every interval is conditional on the fitted models. The classical pipeline is deterministic, seeded and CPU-only; the CNN is not, since 12 of 20 committed grid cells carry dropout $> 0$ and dropout masks are drawn from the device RNG, so CNN results are device-dependent within the five-seed tolerance we report. Because `genomic_benchmarks` 1.0.0 re-extracts on every call we cache locally and pin the cached bytes by SHA-256, recording row counts, extraction timestamps and both file and content digests for all nine cached combinations (Supplementary §S1.8); audits against a different release should re-derive the report card and will see the content digests move.

### S1.8 Cross-suite counting conventions

Two nestings are collapsed in the Nucleotide Transformer census. *enhancers* and *enhancers_types* share identical sequence sets and differ only in label granularity, so they are one task scored twice. *promoter_all* is the exact union of *promoter_no_tata* and *promoter_tata* on both splits, with empty symmetric difference, so it is not independent of them. Collapsing both leaves **eleven independent tasks of the thirteen shipped**. The three splice tasks are *not* nested: *splice_sites_all* is not the union of the acceptor and donor sets, and those two overlap in 35 sequences of roughly 20,000, so they count separately. Although the suite ships original and revised releases we draw no before-and-after inference from the pair: they are not the same data recurated—the original splice tasks are keyed by UniProt accession across species while the revised are human genomic windows keyed by coordinate, and sequence length differs on eight of thirteen tasks—so each release is assessed on its own terms.

For GUE the nesting is analogous. The *_all* promoter tasks' *test* sets are the exact union of their own *notata* and *tata* test sets; the training sets are not, each family withholding a different ≈10% as an unshipped development partition. The 70 bp and 300 bp promoter families are the same loci at two window widths, with 90% of the 70 bp test windows occurring verbatim inside the 300 bp corpus at the fixed offset [216:286]. The independent count is therefore **seven** test partitions: two promoter partitions and five transcription-factor tasks. This nesting is between tasks and affects no verdict, since the census measures test against train within a task, but it bounds how much independent evidence the null carries. These three release-structure statements are re-derived in `results/gue_nesting.csv`.

### S1.9 Software environment

Results were produced with Python 3.13.3, scikit-learn 1.8.0, NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.3 and matplotlib 3.10.9, with the datasets fetched through `genomic_benchmarks` 1.0.0; the from-scratch CNN used PyTorch 2.13.0. Two external tools are called: MMseqs2 `18-8cc5c`

(`--threads 6`) for the alignment-scored re-clustering, and `dustmasker` 1.0.0 (BLAST+ 2.17.0) for the low-complexity check. The classical pipeline is CPU-only, deterministic and hardware-independent. The CNN is *not*: 12 of the 20 committed grid cells carry dropout $> 0$ and dropout masks are drawn from the device RNG, so CNN results are device-dependent within the five-seed tolerance reported in the main text. Reported runtimes and the CNN runs (MPS backend, CPU fallback) are from an Apple M4 under macOS 26.4.1. The pinned dependency list ships as `results/requirements.txt`.

**The audited bytes.** A package version does not by itself identify an input: `genomic_benchmarks` 1.0.0 re-extracts its archives on every call, so a later release, or a re-extraction that orders rows differently, would change what this report card describes without changing anything a reader could see. We therefore pin the bytes. The local extraction was performed on 2026-07-18 (04:53–05:28 UTC) and `results/datacache_manifest.csv` records, for each of the nine cached (dataset, cap, seed) combinations, the row and split counts, the extraction timestamp, and two SHA-256 digests: one over the cached file, and one over the data itself—split, label and sequence for every row in loader order, hashed as UTF-8. The second is the one to compare across machines, being invariant to pickle protocol and pandas version; both are recorded so that a mismatch can be diagnosed rather than merely detected, since file digests differing while content digests agree indicates a serialization change and not a change of data. The two full-scale leaky datasets carry content digests `ebdcf7a65d20f776`... (*human_nontata_promoters*, 36,131 rows) and `50047b21fceb5e3d`... (*human_enhancers_ensembl*, 154,842 rows). Upstream of the package, the reference builds the curators drew on are pinned in `results/reference_provenance.csv`: Ensembl release 97 GRCh38 for *human_nontata_promoters* and *human_enhancers_cohn*, and release 100 for *human_enhancers_ensembl* and *human_ocr_ensembl*, every URL re-resolved on 2026-07-20 with its dated archive host recorded beside it.

## S2 The diagnostic condition $\varphi^*$ and the dose-response experiment

### S2.1 The condition

Stratify a leaky test set by maximum similarity to training into a leaked stratum (similarity $\geq 0.9$, fraction $\varphi$) and a novel stratum ($< 0.5$), and write $h_M$ and $n_M$ for a model's accuracy in each and $g_M = h_M - n_M$ for the graded gap. Then

$$a_M \;\approx\; n_M + \varphi\, g_M, \tag{S2}$$

and for an ordered pair $(A, B)$ with $A$ leading the as-shipped split, writing $\delta = n_B - n_A$ and $\Delta g = g_A - g_B$, the as-shipped margin is $\varphi\Delta g - \delta$, so the benchmark reports the wrong winner when

$$0 < \delta < \varphi\,\Delta g, \qquad \text{equivalently } \varphi > \varphi^* = \delta/\Delta g. \tag{S3}$$

Equation (S2) carries two qualifications. It is an *approximation and not an identity*: the two strata do not partition the test set, since sequences in the intermediate $[0.5, 0.9)$ band belong to neither. That band is 1.4% of the test set on *human_enhancers_ensembl* (maximum reconstruction error 0.002) but 22.4% on *human_nontata_promoters* (maximum error 0.022) and about a third on the Nucleotide Transformer splice tasks, where the approximation error exceeds the margins it would be used to adjudicate. We therefore apply (S3) only where the intermediate band is small and report the band's size wherever we use it. And (S3) is bookkeeping rather than theory—two strata and an ordering—so it is a diagnostic screen, valuable only because $\varphi$, $n$ and $g$ are all measurable on the as-shipped split, and it is not offered as a contribution in its own right. The main text quotes only the dose at which the ranking flips.

### S2.2 The dose-response experiment

We imposed the duplicated-coordinate construction on *human_ocr_ensembl*, which is clean, at ten doses, each downsampled to the control's size and class balance so that arms differ only in near-duplicate content. At each dose we measured $\varphi$, computed $\varphi^*$ for the as-shipped leader against every challenger using only as-shipped novel-stratum accuracies and graded gaps, and then asked whether the ranking actually inverts under the near-duplicate-aware re-split. Prediction **P6**—inversion at every dose with $\varphi > \varphi^*$ and at none below—was committed in the module before it was run.

The condition is right on all ten doses *as single fits*, and fails on replication; both are reported below. As the dose rises the random forest climbs the as-shipped ranking (rank 4 up to dose 0.20, rank 3 at 0.25, rank 1 from 0.30) while remaining last on the corrected split at every dose, including the untouched control. The ranking first inverts at dose 0.30, where $\varphi = 0.106$ against a predicted breakdown point of $\varphi^* = 0.060$, and inverts at all six higher doses; it does not invert at the three lower ones. Here we report the intermediate $[0.5, 0.9)$ band, which runs 1.23–1.52% of the test set across all ten doses, well inside the regime where (S2) is permitted.

**Replication: the threshold does not survive it.** Each dose above is a single fit at one split seed and one model seed, and `ranking_inverts` is a discrete outcome compared between two orderings whose top-two margins are of order 0.005. A three-digit threshold read off one realisation of a Bernoulli outcome is not a measurement, so we replicated the three doses bracketing the transition across a 5 split-seed $\times$ 3 model-seed factorial, 45 trials in all.

The transition is not sharp and the committed threshold does not hold. Inversion probability is 0.60 at $\varphi = 0.088$ (9/15), 0.47 at $\varphi = 0.102$ (7/15) and 0.87 at $\varphi = 0.116$ (13/15). Binomial intervals on the trials are $[0.32, 0.84]$, $[0.21, 0.73]$ and $[0.60, 0.98]$; the correct unit is the split seed rather than the trial, since the split seed drives which rows are duplicated and how clusters are assigned, and clustering on it widens these to $[0.27, 0.93]$, $[0.07, 0.87]$ and $[0.60, 1.00]$, which overlap across all three doses. Split-seed heterogeneity dominates: at $\varphi = 0.088$ the five split seeds invert at rates 0.00, 1.00, 0.33, 1.00 and 0.67.

Two consequences, both against the single-fit result. **P6 is refuted on replication**: it predicted inversion above the diagnostic threshold and *at none below*, and nine of the fifteen trials below the committed $\varphi = 0.106$ inverted. And no three-digit threshold is identifiable in this range at all—the quantity the single-fit sweep reported as 0.106 is better described as a rising probability across $\varphi \in [0.088, 0.116]$ than as a breakpoint. What survives is the monotone tendency of inversion probability with dose, which the $0.60 \rightarrow 0.87$ endpoints support and the non-monotone middle dose does not contradict once the intervals are read (`results/dose_replication_summary.csv`).

**Four limits.** First, two of the ten rows are not fresh trials: `dose_response.csv` rows `dose_0.0000` and `dose_0.9426` are field-identical to `construction_manipulation.csv` rows `A_control_merged_consensus` and `A_manipulated_matched`, differing only in `seconds`, so the experiment supplies eight additional doses beyond the main text's manipulation table rather than ten. Second, three of the ten doses are vacuous, with no admissible challenger pair (`phi_star` empty). Third, the condition can predict "no inversion" two ways—because no challenger has both $\delta > 0$ and $\Delta g > 0$, so no inversion is possible at any leak fraction, or because such a pair exists but $\varphi$ has not reached $\varphi^*$—and *all three* of our negatives are of the first kind, so the sub-threshold arm is never exercised. Fourth, and most sharply: on these ten doses the rule "the ranking inverts exactly when the forest leads the as-shipped split" also scores 10/10, using no novel-stratum accuracies, no graded gaps and no $\varphi^*$ at all. The dose-response shows the condition is *consistent* with a causal manipulation of $\varphi$; it does not show that the condition beats that much simpler rule, because this manipulation confounds the two.

### S2.3 Out-of-sample test on the Nucleotide Transformer suite

Two of the twenty-four cross-suite pairs are informative, and the condition was right on both—but only one is admissible under our own rule. The intermediate band is 0.0% of the test set on *enhancers* and 33.3% on *splice_sites_acceptors*, the latter precisely the regime where the approximation error exceeds the margins being adjudicated, and that pair's as-shipped margin is 0.0032. The correct no-swap call for *splice_sites_acceptors* RF-versus-HGB ($\varphi = 0.077$ against $\varphi^* = 0.257$) is therefore inadmissible and we do not count it. What remains is a single admissible informative call: a predicted swap for *enhancers* RF-versus-HGB ($\varphi = 0.099$ against $\varphi^* = 0.073$), and HistGradientBoosting does overtake the forest after correction (0.887 against 0.869). Every prediction was emitted from as-shipped measurements before any corrected split existed.

Three limits keep this modest. The aggregate score is nearly uninformative: 24 of 24 pair orders were predicted correctly, but a constant "nothing swaps" predictor scores 23 of 24 on the same data, so all but two of the twenty-four opportunities are vacuous by construction. The one swap the condition caught is itself *immaterial* under our own convention, the two models being separated by only 0.005 as shipped—which is why the task reports no inversion. And one admissible informative call, correct, is slender evidence. The reason for the suite-wide null is the condition's own: $\delta \leq 0$ on twenty-two pairs, because the model leading the as-shipped split is, with one exception, also the best model on novel sequences, so there is nothing to invert.

## S3 Alignment-identity control

Re-clustering by MMseqs2 alignment identity and feeding the same whole-cluster splitter reproduces the *human_enhancers_ensembl* effect under an independent clustering engine: the random-forest drop is 0.164 under 8-mer Jaccard and 0.162 under alignment identity, the other models stay near zero under both, and the forest is still demoted to rank 4. The two linkages nearly coincide there because that dataset's cohesion is 0.96. On *human_nontata_promoters* the drop is milder under alignment (0.108 → 0.064; corrected random-forest accuracy 0.820 → 0.864), so the measured effect is partly specific to the 8-mer edge metric—but see §S1.5: MMseqs2 returned a finer partition on that dataset despite a looser identity criterion, so the attenuation is a linkage signature as much as a metric one, and we do not attribute it to the metric alone.

### S3.1 Cluster structure and the low-complexity check

Canonical (reverse-complement-collapsed) $k$-mer features keep the demotion (forest 1 → 4, drop 0.153 on *human_enhancers_ensembl*; 1 → 3, drop 0.097 on *human_nontata_promoters*), so it is not forward-strand overfitting; restricting the corrected test set to GC-in-distribution sequences gives *lower* accuracy (0.787, 0.631), not the inflated original. An MMseqs2 alignment-scored re-clustering reproduces the *human_enhancers_ensembl* effect (0.164 against 0.162) and gives a milder drop on *human_nontata_promoters* (0.108 → 0.064); that arm changes clustering engine and linkage rule together, so the difference is a linkage signature as much as a metric one (Supplementary §S3). Cohesion explains the asymmetry between the two datasets, and geometrically rather than biologically. *human_enhancers_ensembl*'s clusters are 99.6% pairwise with largest-cluster cohesion 0.96—discrete 2× locus duplication that whole-cluster correction removes cleanly—whereas *human_nontata_promoters*'s 420-member largest component has cohesion 0.083. That component is 420/420 *negative* class, non-promoters, and is one locus: 251 bp windows tiling 1,421 bp at a median 2 bp step, 62% of member pairs offset by ≥ 251 bp and sharing no sequence. Its low cohesion is window geometry, not chained paralogy. Dataset-wide, 94.3% of negatives sit in multi-member components against 0.4% of positives, and 2,808 of 2,811 components are class-pure—the same fact §3.2 of the main text reports as $R_{\text{self}} = 0.961$ in the negative class. The certification tool therefore returns *leaky-but-correction-unsafe* rather than *leaky* below cohesion 0.5 and directs the user to the novel-stratum readout instead of a re-split. A `dustmasker` check finds the leaked strata low-complexity-poor (0.033 and 0.053 masked against 0.106 and 0.043 novel), which excludes a microsatellite artifact but is silent on interspersed repeats—DUST masks compositionally biased tracts, and on *human_nontata_promoters* the leaked stratum is in fact ∼7× AluY-enriched relative to novel.

Table S3: Positive-class construction step against measured verdict. **Retrospective**: categories were assigned with the verdicts already known, so the table has zero free parameters but is not a prospective test. The prospective coordinate call is BEND (§3.4); the prospective sequence-space call is GUE (§3.12), which the prediction got wrong. ‡borderline under containment only.

| Dataset | Construction of positives | Merge step | Verdict |
|---|---|---|---|
| h. enhancers_ensembl | un-dedup. assembly | none | LEAKY |
| h. nontata_promoters | fixed window, recurring loci | none | LEAKY |
| h. ocr_ensembl | merged segmentation | MultiCell | clean |
| h. ensembl_regulatory | merged segmentation | MultiCell | clean |
| h. enhancers_cohn | curated, deduplicated | curation | clean |
| d. enhancers_stark | curated, deduplicated | curation | clean |
| demo_coding_vs_interg. | random genome draw | disjoint draw | clean‡ |
| demo_human_or_worm | random genome draw | disjoint draw | clean |

## S4 Detector calibration and the estimator floor

Table S4 gives the sensitivity calibration on synthetic sequences with known ground truth. The instrument is not fixed across length: a 70 bp sequence holds only 63 8-mers, and at that length the Jaccard at 0.7 already misses two point mutations 70% of the time, where at 300 bp and above it tolerates five. It is a near-exact-duplicate detector on short sequences and a homology detector on long ones. Unfloored containment scores 1.0 for a training row of 8 bp, and on *human_enhancers_ensembl* the median best-match training length is 9 bp: under a 100 bp length floor the containment leak fraction reads 0.407 against Jaccard's 0.373, so containment is an *upper* bound on variable-length data, not a lower one, and the two metrics bracket the truth rather than one dominating.

Table S4: Sensitivity of the 8-mer Jaccard at threshold 0.7, on synthetic sequences with known ground truth. Entries are the fraction of perturbed copies still detected. The detector's behaviour changes qualitatively with sequence length; the containment index detects every one of these perturbations at every length shown.

| Perturbation | 70 bp | 200 bp | 300 bp | 500 bp |
|---|---|---|---|---|
| exact copy | 1.00 | 1.00 | 1.00 | 1.00 |
| 1 point mutation | 1.00 | 1.00 | 1.00 | 1.00 |
| 2 point mutations | 0.30 | 1.00 | 1.00 | 1.00 |
| 5 point mutations | 0.00 | 0.00 | 1.00 | 1.00 |
| window shift 10% of length | 1.00 | 1.00 | 1.00 | 1.00 |

**The measured operating characteristic.** Specificity is estimable, and we measure it rather than argue it. Enumeration is not the expensive step—the census already forms every test–train pair by sparse matrix product—adjudication is, so we adjudicate a stratified sample. For each leaky dataset we tally the exact population of pairs in each Jaccard band and reservoir-sample within every band, flagged and unflagged alike: sampling the unflagged bands is what makes recall estimable, since the homologs the detector *missed* live there. Each sampled pair is then adjudicated by Smith–Waterman local alignment (`PairwiseAligner`, EMBOSS `water` scoring), with ground truth the CD-HIT-EST-like criterion the curators omitted: local alignment covering ≥50% of the shorter sequence at ≥80% identity. Horvitz–Thompson weights (band population over band sample) return every rate to the full pair population, and each carries a Clopper–Pearson interval propagated in the direction *worst* for the detector.

Pairs in which either sequence is shorter than 50 bp are excluded and counted separately. *human_enhancers_ensembl* ships rows as short as 2 bp, and on those an alignment ground truth fails in both directions at once: a 6 bp exact match between a 7 bp row and a 300 bp row scores identity 1.00 at coverage 0.86 and would be called homologous though a 6 bp match is expected by chance, while a 15 bp byte-identical pair—a true duplicate at Jaccard 1.0—is called non-homologous by any criterion carrying a minimum alignment length. Both errors are properties of the shipped rows rather than of the detector, and each biases a different rate. The restriction removes 22.0% of sampled *human_enhancers_ensembl* pairs and none of *human_nontata_promoters*'s, which are all 251 bp.

The result is that the flag rule is essentially never wrong when it fires and frequently silent when it should not be. Every flagged band on both datasets adjudicates 100% homologous (3,331 and 4,300 eligible pairs): precision is 1.000, Clopper–Pearson lower bounds 0.951 and 0.978, and the per-pair false-positive rate is 0.000 with upper bounds $1.5 \times 10^{-7}$ and $1.1 \times 10^{-6}$—below the $1/n_{\text{train}}$ scale at which a maximum-over-training-set statistic would be corrupted, which is what the estimator floor needed. Recall is the opposite story and is reported as such: 0.006 on *human_enhancers_ensembl* and 0.260 on *human_nontata_promoters*. The detector misses most true homologs, because homology at 80% identity over half a sequence routinely leaves an 8-mer Jaccard far below 0.7—on *human_enhancers_ensembl* 41.7% of pairs in the $[0.1, 0.3)$ band are genuine homologs, and 94.2% in $[0.3, 0.5)$. This is the quantitative form of the scope the main text already states: the instrument is a near-duplicate detector, not a homology detector, so every leak fraction we report is a *lower* bound on homology-induced leakage, and a homology-aware tool would flag more. Per-band counts are in `results/detector_specificity_measured.csv` and the adjudicated pairs in `results/detector_specificity_pairs.csv`.

**Strand, measured two ways.** The census reads forward-strand $k$-mers, so a reverse-complemented near-duplicate is invisible to it by construction, and a "clean" verdict that has never been measured strand-symmetrically is a weaker statement than the report card implies. Two independent measurements bound the gap. First, hashFrag builds its BLAST database from *both* orientations of every training sequence and labels reverse-strand database entries, so the share of test sequences whose *only* qualifying hit is on the reverse strand is directly readable: at the threshold where the two tools agree on seven of eight verdicts it is 0.0025 on *human_enhancers_ensembl*, 0.0060 on *human_nontata_promoters*, and at most 0.0100 on any of the eight datasets. Second, recomputing the census itself with canonical $k$-mers—each $k$-mer replaced by the lexicographic minimum of itself and its reverse complement—and doing so across the whole suite rather than on the two leaky datasets alone raises every leak fraction slightly and moves *no* verdict on any of the eight, under either metric and at either threshold. The largest Jaccard movement anywhere is 0.0096 (*human_nontata_promoters*, 0.4064 → 0.4160) and the largest containment movement 0.0718 (*human_enhancers_ensembl*, 0.8454 → 0.9172); on the six datasets we call clean the Jaccard movement is at most 0.0030 (`results/canonical_census_suite.csv`). Note that the clean rows there are the same 20,000-sequence subsamples used elsewhere, so they confirm cleanliness rather than establish it. Reverse-strand-only leakage is therefore real but small, and bounded above by roughly one percentage point on every dataset here; the forward-only convention understates leakage by about that much and never by enough to change a verdict.

The maximum-over-training-set caveat survives this and is worth restating: the census statistic is not a per-pair score, so a per-pair false-positive rate must fall well below $1/n_{\text{train}}$ before the resulting fraction is negligible. What has changed is that this is now a measured inequality rather than an argument from construction and provenance. Two further properties are worth recording. Per-pair behaviour on real DNA is much heavier-tailed than on synthetic sequence, because real genomes carry repeats and low-complexity tracts, so a synthetic null would materially understate the floor and the floor argument rests on real clean data rather than on simulation. That tail is heavier than a sample reveals, and a sampled maximum must not be read as a ceiling: on *drosophila_enhancers_stark* a 4,000-pair sample tops out at a Jaccard of 0.169, a 20,000-pair sample at 0.509, and exhaustive enumeration at 0.981.

**The per-pair false-positive rate, measured exhaustively.** We previously reported this rate as 0.000 from 4,000 sampled pairs per clean dataset. That number was not wrong, but it was not evidence: 0/4,000 excludes only rates above $\sim 7 \times 10^{-4}$, while the near-duplicate density in these datasets is $3 \times 10^{-7}$ to $4 \times 10^{-6}$, so the sample would have returned 0.000 whether or not false positives existed. It also rested on a warrant that does not hold—that any similarity between two members of a *clean* dataset is by that verdict not leakage. Clean means below the 0.1 cut, not free of duplicates, and these datasets do contain genuine near-duplicate pairs (83 of them, 27 crossing the split), so a rate computed that way is an upper bound on the false-positive rate rather than the rate itself.

We therefore enumerate rather than sample. Over all 17,997,000 within-dataset pairs of each of the three clean datasets—54 million pairs, no sampling—the 8-mer Jaccard at 0.7 flags 5 pairs on *human_ocr_ensembl*, 78 on *drosophila_enhancers_stark* and 0 on *human_enhancers_cohn*. Adjudicating every one of those 83 by the Smith–Waterman criterion used above returns 83 genuine homologs and *no* false positive: the per-pair false-positive rate is 0.000 with an exact upper bound of $2.05 \times 10^{-7}$, three orders of magnitude tighter than the sampled version and informative at last at the $1/n_{\text{train}}$ scale the maximum statistic requires. Repeat content does not change it. Using `dustmasker` low-complexity fraction and a $k$-mer-reuse score as repeat proxies, repeat-rich pairs sit higher in the body of the distribution (p99 0.110 against 0.019 on *human_ocr_ensembl*) but top out at 0.178, nowhere near the threshold, and the flagged pairs are mostly repeat-*poor*. The proxies are a real limitation and are stated as one: DUST sees simple and low-complexity repeats and is largely blind to interspersed elements, so an AluY is of ordinary composition and invisible to it. This bounds the simple-repeat contribution, not the Alu contribution.

**Indels.** The calibration above covers substitutions and window shifts, and the natural worry is that an indel is worse because it displaces every downstream $k$-mer. It is not, and the reason is that a $k$-mer set is order-free: displaced $k$-mers are still present, merely relocated, so a $d$ bp indel destroys about $k+d$ $k$-mers local to the breakpoint, whereas a substitution destroys $k$ and injects $k$ spurious ones, damaging numerator and denominator at once. At 500 trials per cell, insertions and deletions of 1, 2 and 5 bp are detected at rate 1.000 at every length, and only the 10 bp case at 70 bp falls short—0.576 $[0.531, 0.620]$ for insertions, 0.308 $[0.268, 0.351]$ for deletions—which is a length effect, 10 bp being 14% of that sequence, rather than a displacement effect. Containment detects every indel at every length. The contrast with substitutions is sharp at 70 bp: a 5 bp indel is always detected, while two substitutions are missed 75% of the time. The "near-exact-duplicate detector on short sequences" framing is therefore edit-type-dependent as well as length-dependent (`results/calibration_indels.csv`, `results/calibration_repeats.csv`).

## S5 Out-of-suite coordinate census: BEND

Because the screen needs only shipped coordinates it runs outside the suite, at lengths far beyond anything audited here in sequence space. Applied unchanged to the four supervised tasks of BEND (Marin et al., 2024), whose BED releases carry their own split assignments, it returns 0.000 at $\geq 50\%$ reciprocal overlap on chromatin accessibility (1,410,554 train against 372,153 test intervals), 0.000 on histone modification (433,861 against 120,567) and 0.000 on enhancer annotation in all ten folds—the last at 100,096 bp per sample (Fig. 1B of the main text). Gene

[tbp]

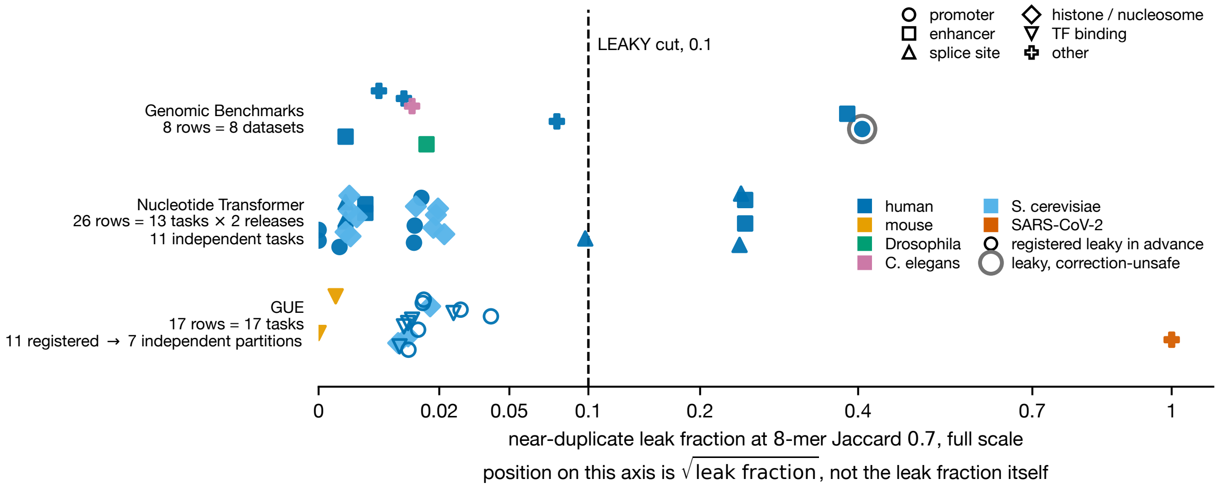


Figure S1: Near-duplicate leak fraction across the fifty-one dataset–task censuses reported here, in three benchmark suites. Each point is one censused dataset–task *row*, which is not the same as an independent task, and the panel annotates both conventions: the Nucleotide Transformer contributes 13 shipped tasks in two releases (26 rows) that collapse to 11 independent tasks, GUE ships 17 tasks of which the 11 registered in advance collapse to 7 independent test partitions, and the text reports independent-task counts throughout. $x$ is the fraction of test sequences whose maximum 8-mer Jaccard to any training sequence exceeds 0.7, measured at full scale; position on the axis is the *square root* of that fraction, as the axis label states, so the sub-0.1 region, which holds forty-four of the fifty-one, is legible. Marker shape denotes task type (13 promoter, 7 enhancer, 6 splice site, 13 histone-mark or nucleosome occupancy, 7 transcription-factor binding, and five singletons: open chromatin, coding-versus-intergenomic, human-versus-worm, three-class regulatory assignment and nine-way viral variant assignment); colour denotes source organism. The dashed rule at 0.1 is the LEAKY verdict cut used throughout. Open symbols are the eleven GUE tasks registered in advance as leaky; all eleven fall left of the cut, at 0.009 to 0.041. The single grey-haloed point is *human_nontata_promoters*, the one dataset certified *leaky-but-correction-unsafe* (Table 1 of the main text). The SARS-CoV-2 point at 1.000 is *virus_covid*, where near-identity is a property of the genome rather than a curation defect (§4). A fourth suite, BEND, is censused in coordinate space only and appears in Fig. 1B of the main text.

finding, the one task partitioned at 80% identity rather than by chromosome, reproduces the *drosophila_enhancers_stark* pattern: 0.183 of its 597 test intervals touch a training interval at $\geq$1 bp, yet only 0.002 reach $\geq$ 50% and none 90%. This is the only prospective construction-based call in the paper, and it is near-tautological on the three chromosome-split partitions. Together with the 7/8 in-suite separation, the screen therefore reads construction defects across four orders of magnitude of interval length without being refitted to the suite it was developed on.

## S6 Manipulation replication, all arms

**Replication: a design, not an anecdote.** With one donor and one recipient the manipulation is $n = 1$ per direction, so we repeated break-a-clean on two further clean donors with nothing retuned—the same target duplication share, rate algebra, arms, split seed and evaluation. The manufactured *inflation* replicates on all three. At matched size and balance, *human_enhancers_cohn* goes from a clean control (forest drop $-0.003$, CI $[-0.021, 0.016]$) to an imposed leak fraction of 0.184 with a drop of $+0.076$ $[0.059, 0.094]$, and *drosophila_enhancers_stark* from $-0.000$ $[-0.032, 0.033]$ to 0.194 and $+0.066$ $[0.030, 0.101]$; both control intervals include zero and both manipulated intervals exclude it. The manufactured *reordering* replicates on two of the three, and the exception is informative rather than awkward: on *drosophila_enhancers_stark* the forest is already rank 4 as shipped, so there is no demotion available for the defect to produce, and it stays rank 4 despite the inflation. Imposing this construction reliably inflates a memorization-prone model; whether that inflation also reorders depends on where the model started.

Separately, the fix-a-leaky direction compared arms of unequal size (154,842 against 81,868), leaving open whether the merged arm's lower score was the merge or simply less training data. A size-matched control—the unmerged assembly downsampled to the manipulated arm's exact 81,868 rows—still carries leakage (0.209) and still inflates the forest ($+0.066$ $[0.055, 0.075]$, rank 2 $\rightarrow$ 4), so the effect is not an artifact of the size difference (`results/construction_manipulation_replication.csv`).

## S7 Robustness arms in full

**The block bootstrap.** That the intervals move little is not because the dependence is absent. The splitter assigns whole components to one side, so every near-duplicate relation surviving correction is test-to-test by construction, and the corrected test sets are the more clustered arm: 43.4% of corrected *human_nontata_promoters* test sequences lie in a multi-member component against 25.2% as shipped, and 47.7% of *human_enhancers_ensembl*'s against 10.1%. Those sizes are heavy-tailed, so Kish's $m^*$ is 4.45 and 2.58 where the mean is 1.55 and 1.33, giving design effects of 2.80 and 2.58 for the forest where the mean-size convention returns 1.29 and 1.33. The inflation is measured directly: on the corrected arm the cluster-bootstrap variance of the forest's accuracy exceeds its sample-bootstrap variance by 3.96× and 2.12× (3.10× and 1.78× for logistic regression), widening the delta intervals 1.2–1.8×. The intraclass correlation is high (0.78–0.99) across the four fits, so the design effect is governed by the block-size distribution rather than by the within-block correlation. The dependence is therefore real, larger than a mean-size design effect suggests, concentrated in the arm the correction creates—and already inside every reported interval, because the block bootstrap resamples those components rather than modelling them. No interval and no verdict changes: the forest's drop still excludes zero on both datasets ($[0.155, 0.173]$; $[0.092, 0.126]$), and Benjamini–Hochberg over the declared 15-delta family leaves the same set significant. One source no test-set resampling reaches is the fit, which is held fixed.

**Chromosome holdout.** The demotion reproduces on *human_enhancers_ensembl*: the forest falls from rank 1 (0.8596) to rank 4 (0.7069), giving the identical corrected order as the near-duplicate-aware split. On *human_nontata_promoters* it does *not*: the forest stays rank 1 (0.9284 → 0.8199), a third independent signal—with the AUROC/$F_1$ reversal and the novel-only rank correlation—that its demotion is accuracy-only. Two disclosures bound that arm: its test set is 74.33% positive against a 47.20% training prevalence, a 27-point shift, and it is scored on accuracy, which the imbalance panel below shows understates the correction. The controls agree on the drop magnitude on *human_enhancers_ensembl* (corrected 0.707 against 0.696); on *human_nontata_promoters* they agree to 0.0001 across test sets 74.3% and 54.4% positive, which is consistent with, but no evidence for, the near-duplicate-aware split proxying the deployment control.

**Metric geometry and cluster structure.** The demotion survives canonical (reverse-complement-collapsed) $k$-mer features (forest 1 → 4, drop 0.153 on *human_enhancers_ensembl*; 1 → 3, drop 0.097 on *human_nontata_promoters*), so it is not forward-strand overfitting, and a GC-in-distribution restriction gives *lower* corrected accuracy (0.787, 0.631), not the inflated original, so it is not a GC covariate shift. An MMseqs2 alignment-scored re-clustering reproduces the effect on *human_enhancers_ensembl* (0.164 against 0.162) and gives a milder drop on *human_nontata_promoters* (0.108 → 0.064); that arm changes clustering engine and linkage rule together, so the difference is a linkage signature as much as a metric one (Supplementary §S3). Cluster cohesion explains the asymmetry between the two datasets, and geometrically rather than biologically: *human_enhancers_ensembl*'s clusters are 99.6% pairwise at cohesion 0.96, discrete 2× locus duplication that whole-cluster correction removes cleanly, whereas *human_nontata_promoters*'s 420-member largest component sits at cohesion 0.083, is 420/420 *negative* class, and is a single locus tiled by 251 bp windows at a median 2 bp step—window geometry, not chained paralogy. That geometry has a documented origin rather than an inferred one, though not the one a reader might guess. The negatives are read from a third-party FASTA that already contains 27,731 uniformly 251 bp records, and the curators' `seq2loc` step only recovers their GRCh38 coordinates by exact string match; it does not create the windows and nothing merges them afterwards. The tiling is visible in the notebook's own printed output, whose consecutive chr1 negatives sit at steps of 23, 8, 41 and 32 bp—eight rows of a preview, so corroboration of our census rather than a census in itself. Dataset-wide 94.3% of its negatives sit in multi-member components against 0.4% of positives and 2,808 of 2,811 components are class-pure, the same fact §3.2 reports as $R_{\mathrm{self}} = 0.961$ in the negative class; the certification tool therefore returns *leaky-but-correction-unsafe* rather than *leaky* below cohesion 0.5. A `dustmasker` check excludes a microsatellite artifact but is silent on interspersed repeats, and on *human_nontata_promoters* the leaked stratum is in fact ∼7× AluY-enriched relative to novel (Supplementary §S3).

**Where the multiplicity family is drawn.** Correcting the per-dataset grid and the cross-suite grid as separate families is more permissive than correcting their union, so we ran the union. Pooling all 31 deltas into one Benjamini–Hochberg family leaves every conclusion in this paper unchanged and adds one flag rather than removing any: *enhancers*|LinearSVC crosses at $q = 0.049$. Pooling is not the one-sided stress test it looks like—enlarging a family raises the number of tests, which is restrictive, but it also raises the rank of each $p$-value when the incoming tests are more significant than the incumbents, and the per-dataset family contributes several $p$-values below $10^{-5}$. Both effects are real and the analysis has to be run rather than reasoned about (`results/bh_pooled.csv`).

**Against a published tool.** hashFrag (Rafi et al., 2026) is the closest runnable comparator: BLAST-based, alignment-scored, and strand-explicit, building its database from both orientations of every training sequence. Run on all eight datasets, it agrees on the two verdicts that matter and disagrees in an instructive way about the rest. Its threshold is the whole story. hashFrag supplies no default and its documentation recommends scoring dinucleotide-shuffled sequences and cutting above that null; done per dataset, that procedure returns thresholds of 21–88 and calls *every* dataset leaky, including all six we call clean. That is the failure our own supplement predicts on independent grounds—real genomic DNA carries repeats and low-complexity tracts that shuffling destroys, so a synthetic null puts the cut far below the scale of real alignment scores—and it is a caution about the recommended procedure rather than about the tool. Sweeping the threshold instead, a window at 220–240 reproduces 7 of our 8 verdicts exactly, with per-sequence agreement of 0.837 on *human_enhancers_ensembl* and 0.958 on *human_nontata_promoters* and Cohen's $\kappa$ of 0.636 and 0.913 respectively. The separation is in the right direction and is not an artifact of choosing that window: the two leaky datasets require thresholds of 260 and 380 before their leak fractions fall under the verdict cut, against 100–220 for the clean ones.

The single persistent disagreement is *drosophila_enhancers_stark*, which hashFrag calls leaky at every threshold we swept while we call it clean, and it marks the real boundary of our instrument rather than an error in either tool. Its sequences are long (2,142 bp median) and tile the genome, so neighbouring tiles share flanking sequence over hundreds of bases: BLAST scores those alignments highly, while an 8-mer Jaccard is bounded by the length ratio and stays low. The dataset is clean of near-duplication and is not clean of homology—the same distinction the measured recall of §4 quantifies, arrived at through a second tool and a different algorithm. Symmetrically, a single absolute alignment score is itself length-dependent: at threshold 220 hashFrag reads *lower* than we do on *human_enhancers_ensembl* (0.271 against 0.384), because that dataset's short sequences cannot reach 220 at all. Neither instrument is threshold-free, and both should have their cut follow sequence length.

**Imbalance, and what "corrected" means.** Under induced prevalence imbalance the inflation is revealed by prevalence-aware metrics and masked by accuracy: on *human_enhancers_ensembl* at $\pi = 0.2$ correction lowers AUPRC 0.622 $\to$ 0.477, MCC 0.422 $\to$ 0.182 and minority recall 0.258 $\to$ 0.070 while accuracy barely moves (0.844 $\to$ 0.808), and on *human_nontata_promoters* AUPRC falls 0.949 $\to$ 0.813 and MCC 0.785 $\to$ 0.682 against accuracy's 0.934 $\to$ 0.903—but minority recall is flat there (0.673 $\to$ 0.674), so the recall collapse does not replicate. What generalizes is that AUPRC and MCC fall several times further than accuracy on both datasets, so accuracy at threshold 0.5 understates the correction; the panel is random-forest-only and its prevalence induced rather than native, leaving the reordering claim untested under imbalance. Across the four robustness arms—block bootstrap, chromosome holdout, metric geometry and induced imbalance—no interval and no verdict changes, and the dependence the block bootstrap measures is real, larger than a mean-size design effect suggests, and already inside every reported interval.

## S8 Imbalanced evaluation: full panel

Table S5 gives the prevalence panel at $\pi = 0.2$. What generalizes across both datasets is that AUPRC and MCC fall several times further than accuracy—by 0.145 and 0.241 on *human_enhancers_ensembl* against accuracy's 0.036, and by 0.136 and 0.103 on *human_nontata_promoters* against accuracy's 0.030—so accuracy understates the correction on both. The minority-recall collapse does *not* generalize: it is specific to *human_enhancers_ensembl* (0.258 $\to$ 0.070) and flat on *human_nontata_promoters* (0.673 $\to$ 0.674), where minority recall (0.001) and balanced accuracy (0.019) both move less than accuracy (0.030).

Table S5: Prevalence stress test at $\pi = 0.2$, random forest, as-shipped $\rightarrow$ corrected. The panel is random-forest-only and the prevalence is induced by downsampling both train and test, not native, so it does not test the reordering claim under imbalance.

| Metric | *h. enhancers_ensembl* | *h. nontata_promoters* |
|---|---|---|
| Accuracy | $0.844 \rightarrow 0.808$ | $0.934 \rightarrow 0.903$ |
| AUPRC | $0.622 \rightarrow 0.477$ | $0.949 \rightarrow 0.813$ |
| MCC | $0.422 \rightarrow 0.182$ | $0.785 \rightarrow 0.682$ |
| Minority recall | $0.258 \rightarrow 0.070$ | $0.673 \rightarrow 0.674$ |

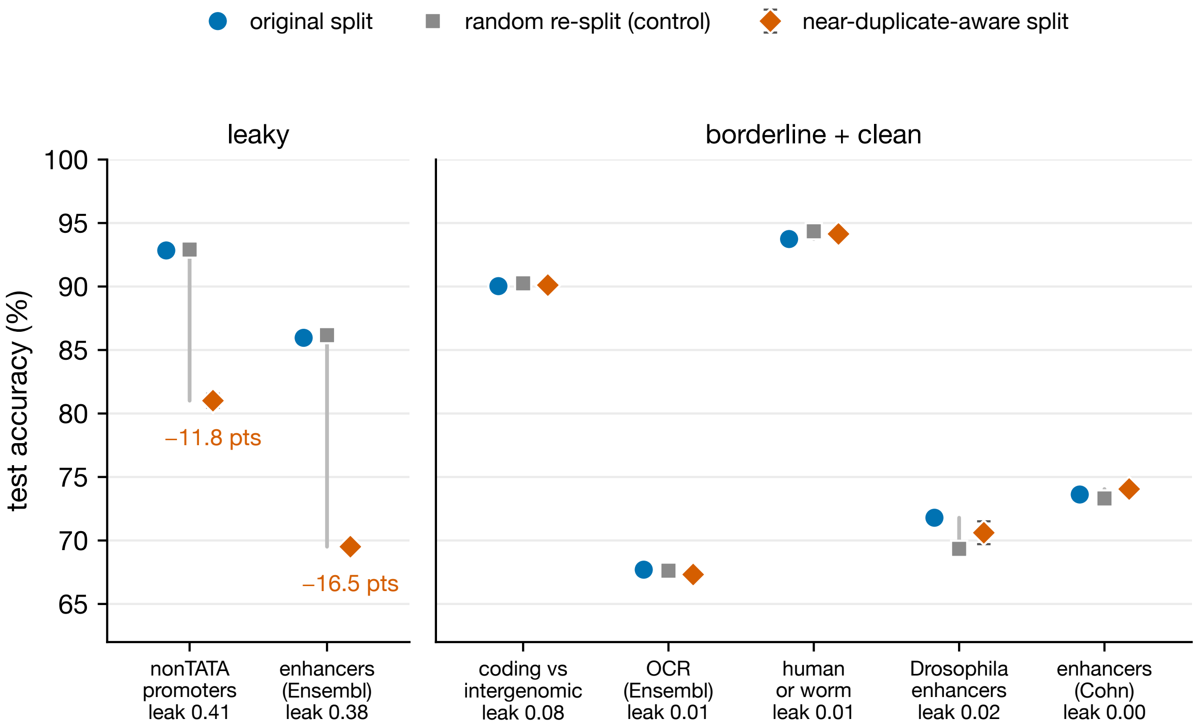


Figure S2: Original split versus random re-split (negative control) versus near-duplicate-aware re-split, best model per dataset, for all seven binary datasets. On the two leaky datasets (left) the near-duplicate-aware split costs substantial accuracy while the matched random re-split does not. On the borderline and clean datasets (right) both deltas are null—the largest, *drosophila_enhancers_stark*, moves +0.012 under the near-duplicate-aware split against +0.025 under the random control. That the control sometimes moves a clean dataset further than the near-duplicate-aware split is the expected behaviour of a null. The best model is the random forest on the two leaky datasets and logistic regression on the other five, both at $k$=6; "leak" is the near-duplicate test fraction at Jaccard 0.7; error bars on the corrected points are the standard deviation over re-split seeds. The random-control deltas are computed on the pre-reconciliation decision rule and are not directly differenced against the corrected values.

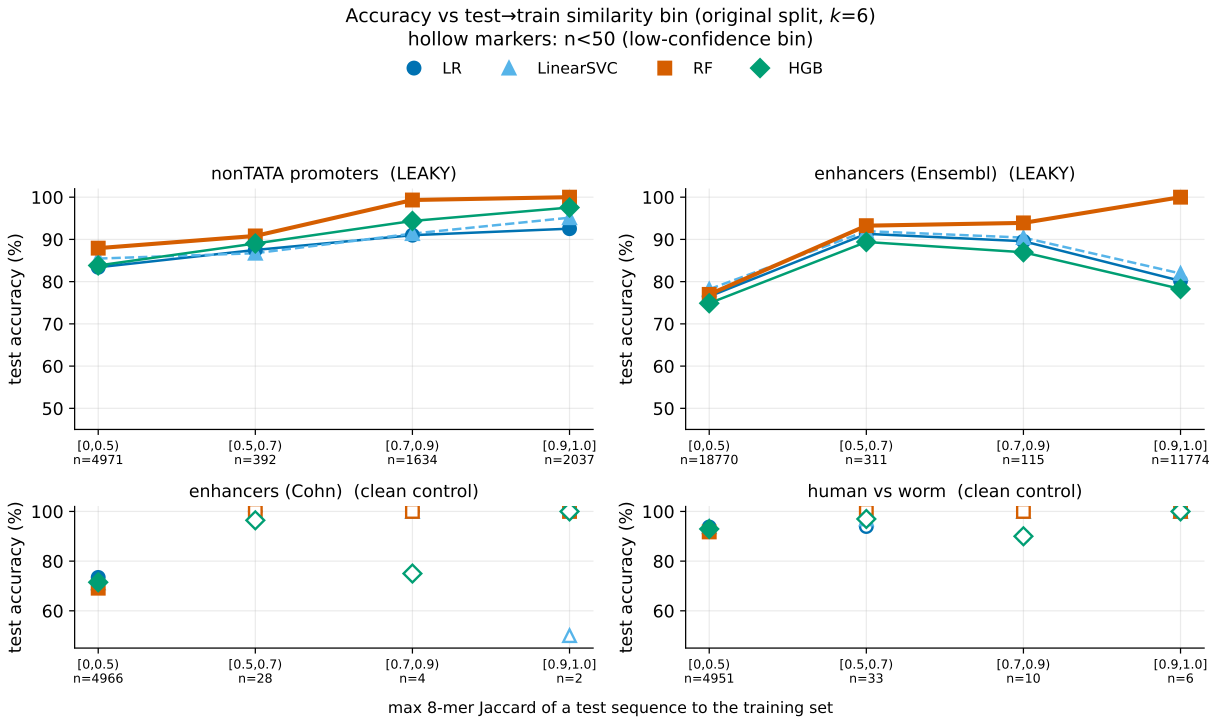


Figure S3: Accuracy as a function of the test-to-train similarity bin (as-shipped split, $k$=6, the four-model comparison). On leaky datasets the near-default random forest is near-perfect on high-similarity (near-duplicate) test sequences and falls on dissimilar ones; clean controls have near-empty high-similarity bins. Hollow markers denote bins with $n < 50$.

# S9 Extended results: controls, grids and the injected multiclass arm

**Regularization path (`results/exp_regpath.csv`).** Twenty-two rows sweeping `min_samples_leaf`$\in \{1, 5, 20, 50, 100, 200, 500\}$ and `max_depth`$\in \{\text{None}, 16, 8, 4\}$ at full scale on both leaky datasets, reporting per cell the as-shipped and corrected accuracy, the drop, the graded gap and the model's rank on each split. The endpoints quoted in the main text are its first and last leaf-size rows. Note the single 0.0012 non-monotonicity in the last *human_nontata_promoters* leaf-size step, and that regularization lowers corrected accuracy on that dataset (0.820 $\rightarrow$ 0.785) while removing the inflation, so the two interventions are not interchangeable.

**CNN grid (`results/exp_deep_cnn.csv`).** Twenty cells: nine grid cells plus one unregularized-to-convergence manipulation check per dataset, each with as-shipped accuracy, corrected accuracy, drop, cluster-bootstrap interval and graded gap. Nineteen of the twenty drop with intervals excluding zero. The sole exception is the *human_nontata_promoters* dropout-0.6, weight-decay-$10^{-4}$ cell, where accuracy rises 0.020 after correction; we report that anomaly bare rather than explaining it, since that cell's graded gap of 0.284 is the largest of the ten on the dataset and is not consistent with a simple under-fitting account. The unregularized-to-convergence check memorizes hardest on *human_enhancers_ensembl* (graded gap +0.222, drop +0.076 [0.068, 0.084]); on *human_nontata_promoters* that same cell's graded gap is 0.075, second-lowest of ten, so the "memorizes hardest" reading is specific to the first dataset. Each cell is a single run.

**Injected-leakage multiclass arm.** The classical arm (`results/exp_inject3class.csv`) is one run per model per dose; the deep arm (`results/exp_inject3class_deep.csv`) is three seeds per dose, Table S6. Seed 1 underfits throughout, with as-shipped accuracy averaging 0.6856 against 0.7552 and 0.7534 for the other two seeds; it is retained rather than excluded.

Table S6: Injected-leakage multiclass, deep arm, on *human_ensembl_regulatory*. *Drop* is as-shipped minus corrected accuracy, so a negative value means correction *raised* accuracy. Here $f$ is the fraction of training rows copied into the test set, giving realised test-set leak fractions $\varphi = 0.29$, 0.44 and 0.62 at $f = 0.1$, 0.2 and 0.4. The random forest on the identical construction drops +0.0004, +0.1177, +0.1791 and +0.2581.

| CNN drop | $f = 0.0$ | $f = 0.1$ | $f = 0.2$ | $f = 0.4$ |
|---|---|---|---|---|
| seed 0 | −0.0284 | −0.0023 | −0.0160 | +0.0064 |
| seed 1 | −0.0659 | −0.0524 | −0.0609 | −0.0502 |
| seed 2 | −0.0216 | −0.0000 | +0.0023 | +0.0172 |
| mean | −0.0386 | −0.0182 | −0.0249 | −0.0089 |
| sd | 0.0239 | 0.0296 | 0.0325 | 0.0362 |

There is no dose-response: every mean is negative, and the seed standard deviation equals or exceeds any dose effect. This is the outcome "memorization propensity, not capacity" predicts, and it is the arm that scopes the multiclass generalization claim to the classical memorizer.

## S10 Exploratory analyses

Everything in this section is exploratory. The prevalence-corrected graded gap was defined after P1 had been scored on its registered raw form, and the curve-shape comparison was not registered at all; neither can confirm the hypothesis it bears on.

**Exploratory: the prevalence-corrected gap.** Everything from here to the end of this subsection is exploratory and is reported as such. P1 was committed against the *raw* graded gap, and the prevalence-corrected form below was defined after that prediction had been scored, so it cannot confirm the hypothesis it was built to rescue—it can only motivate the prospective test we register at the end of this paragraph. Correcting the class-composition confound *strengthens* the mechanism: from balanced accuracy, 1-nearest-neighbour, the definitional memorizer, has by far the largest gap (+0.461), then extremely randomized trees (+0.370) and the forest (+0.319), with the linear models (+0.154, +0.144), naive Bayes (+0.129) and, lowest, 15-nearest-neighbour (+0.016)—averaging over fifteen neighbours is what stops a nearest-neighbour rule memorizing. The four memorization-prone learners average +0.292 against +0.165, a separation the raw statistic does not produce. Memorization pays because near-duplicates carry the label: at $\geq 0.9$ similarity 99.99% of near-duplicate test sequences share their nearest training neighbour's label on *human_enhancers_ensembl* ($n = 11{,}774$) and 99.95% on *human_nontata_promoters* ($n = 2{,}037$); clean datasets have near-empty high-similarity bins (Supplementary Fig. S2). The correction is equally informative where it removes a signal: on *human_nontata_promoters* the corrected gaps do *not* separate the families—memorizers average +0.122 against +0.179 for the other five, the largest belonging to the MLP—so the mechanism claim is scoped to *human_enhancers_ensembl*. We therefore register the corrected gap forward rather than counting it backward: on the next leaky dataset audited, 1-nearest-neighbour will carry the largest prevalence-corrected graded gap and the four memorization-prone learners will average a larger corrected gap than the remaining five. That prediction is falsifiable on one dataset, and *human_nontata_promoters* above is the case in which it would already have failed.

**Exploratory: the shape of the curve, against a published one.** That accuracy grades with training-set similarity, and that models rely on memorized associations where it does, is established by Rafi et al. (2026) across three modelling regimes; it is not claimed as novel here, and what this suite adds is a matched clean control and an editable construction step. Their curve is non-monotonic—best at both extremes of homology, worse at intermediate similarity—so ours is worth reporting whether or not it agrees. Binning test sequences by maximum similarity to training and reading *balanced* accuracy, so the prevalence confound above cannot manufacture a shape, the two suites disagree in an interpretable way. The memorizing forest is monotone increasing on both leaky datasets (0.681 → 0.818 → 0.910 → 1.000 on *human_enhancers_ensembl*; 0.788 → 0.922 → 0.997 → 1.000 on *human_nontata_promoters*), which is what a memorization account predicts and what their non-monotonic curve does not show. The models that do *not* memorize are non-monotonic on both, but in opposite directions: on *human_nontata_promoters* logistic regression dips at intermediate similarity (0.824 → 0.755 → 0.800 → 0.963), the shape they report, while on *human_enhancers_ensembl* it peaks there instead (0.747 → 0.955 → 0.886 → 0.901). Two things limit this. The intermediate bins are small on *human_enhancers_ensembl*—311 and 115 sequences against 18,770 and 11,774 at the extremes—so the peak rests on few observations; and their regimes reach 196 kb contexts and functional divergence between homologs, neither of which a $k$-mer model on ≤573 bp windows can express. We report the difference rather than resolve it (`results/graded_shape_summary.csv`).

## S11 An overlap-free protocol for pretrained models

Two routes would settle whether deployed foundation models inherit this inflation, and neither is performed here. The first uses the stratum HyenaDNA provably never saw: it inherits the Enformer interval set and designates chromosomes 14 and X as held-out pretraining test chromosomes, and both are present in our data, carrying 6.6% of *human_enhancers_ensembl*'s test set and 15.3% of *human_nontata_promoters*'s—the latter unevenly by class (24.6% of test negatives against 7.5% of positives, which is itself a confound to handle). The second applies break-a-clean to *drosophila_enhancers_stark* and fine-tunes on both arms under both splits, manufacturing the contamination inside a genome the model has never seen: 6,914 sequences of at most 3,237 bp and twelve runs of `hyenadna-small-32k`. Both are stated so the gap is priced rather than merely disclaimed.

## S12 The Nucleotide Transformer splice mechanism

The splice mechanism is not the cross-species homology an earlier account proposed. Of the leaked test sequences, 65.9% (acceptors) and 68.3% (donors) share the same UniProt entry name with their best training partner—same gene, same organism—and at Jaccard ≥ 0.9 that share *rises* to 77.1% and 82.6%, where a homology account predicts it should thin; same-species-different-gene pairs number four of 360 and four of 372, so it is not paralogy either. The construction draws each gene's decoy negatives from that gene's own exon, intron and false-positive regions, so redundant windows over one gene span the split; label concordance among leaked pairs is correspondingly poor, 0.819 and 0.792 against 0.999 on both Genomic Benchmarks datasets. That account is offered after the fact, not predicted. Same-species-but-different-gene pairs number four of 360 and four of 372, so it is not paralogy; the window-type pairing among same-gene pairs (exon-to-exon, false-positive-to-site, intron-to-site) matches the Spliceator/G3PO+ construction, which draws each gene's decoy negatives from that gene's own exon, intron and false-positive regions. Decomposed, label concordance is 0.725 and 0.701 among same-gene pairs against 1.000 and 0.988 among cross-gene pairs, so memorizing a near-duplicate is wrong roughly a fifth to a quarter of the time on these tasks.

## S13 Use of AI and large language models

Generative AI assistants were used for code drafting, for editorial revision of manuscript prose, and for adversarial review of the manuscript's claims against the released CSV artifacts. They were not used to generate, select, or interpret any experimental result: every number reported in the manuscript and in this supplement is produced by a committed script in the released repository and is re-derived from the corresponding CSV by `audit/tools/check_numbers.py`. The author takes full responsibility for the content of the submission.

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